

Tabletop Robot for Vision-Based Medication Verification at Home

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Abstract. Older adults who manage their own medication at home make frequent errors. Polypharmacy, cognitive decline, and physical limits all play a part. Most assistive technology only reminds people to take a dose. It does not check that the pills in hand are the right ones. A review of 25 papers shows that no prior study tests vision-based pill counting or verification on a real home robot. This project asks whether a tabletop social robot (Reachy Mini) with a vision-language model (VLM) can help with that verification step. We first tested three VLMs on 15 controlled pill images using three prompts. Gemini was the most accurate. Overlapping and touching pills were the main failure mode. Prompting on its own did not fix counting errors. We then built a voice-conversational verification system. It scans bottles, verifies doses, sets reminders, and checks dose history. It also has safety rules that keep the robot inside the trust boundary older adults prefer: it verifies and reminds, but it never gives clinical advice. A four-route camera test (20 trials) shows that the system counts reliably except when pills are dense and overlapping. That limit comes from the vision task itself, not from the robot’s camera or code path. We show three end-to-end interactions and discuss the design, its limits, and future work.

1 Introduction

Managing medication at home is one of the hardest parts of aging in place. Many older adults take several medications on different schedules. This is called polypharmacy. Cognitive decline makes it worse, especially decline in prospective memory, which is the ability to remember to do something later [11, 21]. Physical limits like reduced dexterity and poor vision also make it harder to handle pills and read labels [3]. The most common error is a missed dose. About 18% of home-dwelling older adults miss at least two doses per week, and 23% report trouble remembering to take their medications [4]. Adherence depends heavily on how much support a person has. It runs from about 25% in institutionalized patients with little help to nearly 98% in pharmacist-led programs [5]. Non-adherence is estimated to cost about \$100 billion per year in the United States [19].

A lot of technology tries to help. This includes electronic pillboxes, smartphone apps, smart dispensers, and dispensing robots. Most of it focuses on reminding the user to take a dose. Very little of it checks that the pills about to be taken are the right ones. Section 2 goes through a review of 25 papers. Computer vision for pill identification and counting shows up in only a few of them, and none test those abilities on a real home robot. This project focuses on that gap.

The main question is simple. Can a low-cost tabletop social robot, using a general-purpose vision-language model, reliably support medication verification at home? This paper makes the following contributions:

1. A controlled benchmark of three VLMs (ChatGPT, Claude, Gemini) on pill counting across 15 images and three prompts (Section 3).
2. A prompting study that tests whether prompt engi-

neering can fix VLM counting errors (Section 4).

3. A working voice-conversational verification system on Reachy Mini. Its safety rules follow the trust preferences reported in the literature (Section 5).
4. A four-route camera comparison test that shows where counts break in the full system (Section 6).
5. Three demonstrated end-to-end interactions (Section 7).

2 Related Work

Pillboxes and organizers are the most studied category. One review catalogued 17 electronic pillbox systems, from simple organizers to devices with alarms and locking. It found that many are poorly designed for older adults [6]. **Smart dispensing systems** add real-time tracking and cloud syncing. A scoping review found 114 smart adherence products, about 70% of them already on the market [7]. **Robotic dispensers** are the closest prior work. The Evondos E300 is a Class I medical device in the EU. It hands out pre-packaged sachets on schedule and reached 98.7% on-time dose delivery in homes [4]. It works well, but it only handles pre-packaged medication and does not verify loose pills. **Apps** such as ALICE let patients view images of their medications and get interaction warnings. In that study 88% of users reported feeling more independent [8]. **Ambient displays** use indirect cues instead of explicit alarms. One study raised adherence from 81% to 96% during the intervention. It then dropped to 77% after the technology was removed [10]. That drop is a warning about technology dependence. **AI methods** include reinforcement-learning reminder personalization, deep networks with OCR for container identification, and NLP for drug-drug interaction detection [9].

Table 1: Experiment 1: pill-counting accuracy (% correct, $n=15$) and mean absolute error (MAE, pills) by model and prompt. P1 = naive, P2 = careful, P3 = double-check.

Model	Accuracy			MAE	
	P1	P2	P3	Overall	Overall
ChatGPT	60%	53%	73%	62%	0.44
Claude	73%	73%	73%	73%	0.51
Gemini	93%	100%	93%	96%	0.04

Two findings guided the design. First, trust is task-specific. Older adults are fine with robots handling reminders, but they want humans to make medication decisions [18]. Second, the best systems share a few design principles. They use multi-modal cues (audio and visual), large fonts, caregiver integration, fail-safe escalation, and personalization that respects existing routines [4, 6, 21]. We found no study that tests vision-based pill counting or color-based verification on a home robot. That is the contribution of this work.

3 Experiment 1: VLM Model Comparison

Goal. Find which general-purpose VLM counts pills most reliably, and find the conditions that make counting fail.

Method. We built 15 pill images. The true counts were 3, 5, 7, 10, and 15. The images covered a set of difficulty variables: baseline, spacing, touching, overlapping, background clutter, lighting, shadows, color contrast, and color uniformity. We sent each image to three models: OpenAI ChatGPT, Anthropic Claude, and Google Gemini. We used three prompts for each one. The first was a naive prompt (“How many pills are in this image?”). The second was a careful chain-of-thought prompt (“count each visible pill systematically”). The third was a self-verification prompt (“after you answer, double-check by counting again and confirm or correct”). For each trial we recorded the model’s count, the signed error, and whether it matched ground truth.

Results. Table 1 shows accuracy and mean absolute error (MAE) by model and prompt. Gemini was the clear winner. It was near-perfect across all three prompts, and its only errors were on overlapping pills. Claude did well on separated pills but undercounted badly when pills touched or overlapped (off by 3 to 4). ChatGPT got worse at high counts (15) and on low-contrast images. Background and lighting mattered much less than expected.

Key finding. Two things drive counting reliability: the choice of model and the physical arrangement of the pills, mainly whether they touch or overlap. Surface image properties matter less. This is why we used Gemini as the production backend (Section 5).

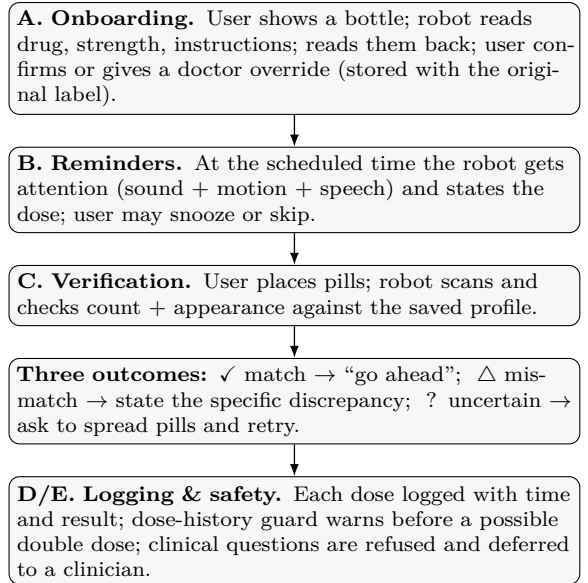


Figure 1: Medication workflow. The robot verifies and reminds but never makes a medication decision, keeping it inside the trust boundary older adults prefer [18].

4 Experiment 2: Does Prompting Fix Errors?

Once we knew Gemini was the best model, we tested whether prompt engineering could push accuracy higher. We ran ten prompting strategies on Gemini. They ranged from a naive baseline to color-first decomposition, bounding-box detection, spatial partitioning, an uncertainty-aware medication-safety persona, and a Set-of-Mark-style numbered-labeling prompt.

The result was mostly negative. Prompting did not reliably fix counting errors. When a model miscounted, asking it to “double-check” or to “go through systematically” usually gave the same wrong number again. The reasoning was more structured and more confident, but the answer did not change. The one promising direction in our notes was decomposition by color subgroup. Breaking the count into smaller groups of one color seems to reduce estimation error. The practical takeaway is that better hard-case accuracy depends more on what the model sees, such as capture quality and spreading the pills out, than on how it is asked.

5 System Design

Architecture. The system builds on a conversational Reachy Mini application. A real-time voice model handles speech. The robot’s head-mounted camera provides vision. Gemini does all label reading and pill counting. A lightweight in-memory store holds the saved medications and a dose log. A web UI mirrors the interaction. Figure 1 shows the medication workflow in five phases.

Design choices from the literature. The system follows the design principles from Section 2. (i) It uses

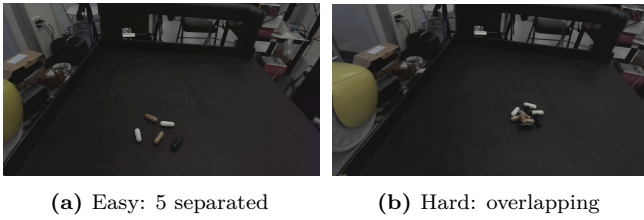


Figure 2: Reachy Mini’s camera view. Separated pills (a) are counted reliably across all routes; clustered, overlapping pills (b) are where every miscount occurred.

multi-modal cues (speech, head motion, and a screen) for reminders. (ii) It has a fail-safe three-way verification outcome, so the robot never confidently confirms a bad count. (iii) It handles doctor overrides directly. It stores the change next to the original printed label and flags it. (iv) It keeps trust-aware task boundaries. The robot reads labels, counts pills, compares against saved profiles, and reminds the user. It is not allowed to give medical advice, recommend dosage changes, or answer interaction questions. A dose-history check warns the user when a medication is already logged as taken. This guards against an accidental double dose.

One important engineering decision. An early version of the code used whichever VLM API key was present. In practice that selected Claude, the worst performer on overlapping pills in Experiment 1. We forced the backend to Gemini, the best model in our tests. This was a one-line change, but it had a large effect on real-world accuracy. It also shows why running the benchmark before the build mattered.

6 End-to-End Evaluation: Camera Comparison Test

Goal. Clean photos counted well in Experiment 1, but the robot still miscounted sometimes in daily use. We designed a test to find where the error comes from: the model, the captured photo, or the code path.

Method. We used four pill setups: Easy, Medium, Hard, and a denser Hard-Closer / Hard-No-Overlap variant. For each setup we ran the same arrangement through four routes. Route (A) was a clean phone photo sent straight to Gemini. Route (B) was that same photo sent through the robot’s own code path. Route (C) was the robot’s three captured snapshots sent to Gemini. Route (D) was the full robot scan from start to finish. We kept the arrangement fixed across routes, so any change in the answer points to one stage. Figure 2 shows the robot’s view of an easy and a hard arrangement.

Results. The returned count matched ground truth in 14 of 20 trials. Every mismatch happened in a dense or overlapping layout (the Hard and Hard-Closer setups). The undercounts were 1 to 2 pills. The largest miss was a full scan that returned 8 instead of 10. The Easy, Medium, and Hard-No-Overlap setups matched across

Table 2: End-to-end camera comparison test, by setup (4 routes each). Misses occur only when pills overlap.

Setup	Matches / 4	Where it failed
Easy (5, separated)	4/4	none
Medium (7, some touching)	4/4	none
Hard (10, overlap)	2/4	robot routes (C, D)
Hard-Closer (10, dense)	0/4	all routes
Hard-No-Overlap (10)	4/4	none
Total	14/20	dense/overlap only

all four routes (Table 2).

Interpretation. Two patterns stand out. On moderate overlap (Hard), the clean phone-photo routes (A, B) counted correctly, but the robot’s own captures (C, D) undercounted by one. This points to a small capture-quality effect. At extreme density (Hard-Closer), every route failed, including the clean phone photos. That is an inherent limit of counting occluded pills, and it is the same failure mode from Experiment 1. The code wrapper (B) never changed the answer compared to a direct model call (A), so the pipeline is not the source of error. These results point the next version toward a few fixes: de-cluttering the scene by asking the user to spread the pills out, higher-quality and multi-angle capture, and color-subgroup decomposition. Swapping models or rewriting the pipeline is not the priority.

7 Demonstration

We filmed three end-to-end interactions. *Demo 1 (label, save, remind):* the user shows a bottle. The robot reads the drug name, strength, and instructions, then reads them back. It asks for confirmation or a doctor override, saves the medication, and offers to set a reminder. The reminder then fires. *Demo 2 (verify and history guard):* the user asks the robot to confirm that a pill is their lisinopril. The robot confirms the match and logs the dose when the user takes it. Later the user starts to take it again, and the robot warns that it was already logged earlier that day. *Demo 3 (wrong count):* the user has a saved dose of a known size and presents a tray with the wrong number of pills. The robot refuses to confirm and states the specific problem, for example one extra pill. A fourth scripted interaction tests the safety rule. When asked an interaction question, the robot declines and points the user to a doctor or pharmacist.

8 Discussion

The system works well for the cases that matter most in everyday verification. It reads a label, saves a regimen with a doctor override, confirms a correctly sized dose, detects a wrong count, and prevents a double dose. Its one weakness is narrow and well understood: overlapping pills. That weakness is a property of the vision task, not of this particular robot. The architecture also stays

within an appropriate trust boundary. It verifies and reminds but refuses clinical judgments, which matches the split of work older adults say they prefer [18]. We also keep in mind the technology-dependence concern from the ambient-display study, where adherence gains disappeared after the device was removed [10]. An always-present home robot may help with this, but it does not teach lasting habits on its own.

9 Limitations

This is a single-developer prototype tested by its author, not a clinical study. There were no real older-adult participants and no IRB-approved evaluation, and the lighting was lab-controlled. The medication library and dose log are in-memory and single-user. We tested counts on capsule-style pills against known ground truth. Tablets, split pills, and real cluttered home settings are still untested.

10 Future Work

There are several next steps. The most important is a pilot with real older adults and caregivers. Others include a caregiver portal with escalation for missed doses and repeated failures, persistent multi-user storage, and drug-drug interaction checks that are routed to a clinician instead of answered by the robot. We would also like to add on-device or privacy-preserving vision and full scheduling with missed-dose policies. Finally, the overlap failure mode deserves direct work through guided de-cluttering, multi-angle fusion, and color-subgroup decomposition.

11 Conclusion

A low-cost tabletop robot with a general-purpose VLM can support home medication verification. Prior robotic medication systems dispense and remind, but they do not check loose pills. Our results show that accuracy depends on the choice of model (Gemini) and on pill arrangement, not on prompt wording. The system’s one real weakness is overlapping pills. That weakness comes from the vision task itself and can be addressed through better capture rather than a different model. Together with safety rules that respect the trust boundary older adults prefer, this is a credible first step toward vision-based medication verification in the home.

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